

# Candidate Profile: Brij Patel

**Role Identity:** Machine Learning Engineer, MLOps Engineer, AI Engineer

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## Professional Summary

Final-year Computer Science and Engineering student at IIIT Nagpur specializing in Machine Learning Engineering, Machine Learning Operations (MLOps), and Large Language Models (LLMs). Proven track record of building production-grade Retrieval-Augmented Generation (RAG) systems, designing cloud architectures, and delivering scalable AI microservices with explicit latency and cost engineering.

## Education

- **Institution:** Indian Institute of Information Technology, Nagpur (IIIT Nagpur)
- **Degree:** Bachelor of Technology (B.Tech) in Computer Science and Engineering
- **Timeline:** 2022 to 2026
- **Academic Performance:** Cumulative GPA of 7.7 out of 10.0

## Professional Work Experience

- **Company:** NexureTech
- **Role:** Artificial Intelligence (AI) Intern
- **Timeline:** January 2026 to March 2026
- **Key Accomplishment 1:** Architected and deployed a production-ready Retrieval-Augmented Generation (RAG) system capable of processing and querying a corpus of over 56,000 documents. Utilized LangChain for orchestration and Qdrant as the high-density vector database backend.
- **Key Accomplishment 2:** Engineered concurrent data ingestion pipelines that optimized resource utilization and improved document ingestion throughput by 20x compared to legacy pipelines.
- **Key Accomplishment 3:** Productionized and hardened Large Language Model (LLM) inference pipelines by implementing rigid Pydantic schemas for structured type validation, asynchronous I/O ([asyncio](#)) to prevent blocking calls, and robust automated exponential-backoff retry logic.

## Core Engineering Projects

- **Project 1: Production MLOps Pipeline on AWS EKS**
  - **Details:** Deployed highly scalable, production-grade machine learning microservices on AWS Elastic Kubernetes Service (EKS), configuring self-healing nodes and load balancing to sustain a 99.9% application uptime.

- **CI/CD:** Authored custom GitHub Actions workflows that automated the build, test, and container deployment pipeline, dropping total manual deployment times by 80%.
- **Project 2: Multimodal Temporal Emotion Recognition System**
  - **Details:** Devised a tri-modal neural architecture fusing text, audio, and video streams to classify complex emotional states, hitting a 70 to 75% Weighted F1-score on the standard MELD dataset benchmarking suite.
  - **Optimization:** Precomputed and cached foundational model embeddings to drop aggregate pipeline training latency by 60%.
- **Project 3: LLM Resume Parsing & Extraction System**
  - **Details:** Engineered an end-to-end extraction pipeline to pull structured JSON from inconsistent resume inputs. Implemented a hybrid preprocessing step ("Hybrid Shield") combining text extraction and structural stripping to decrease OpenAI LLM token consumption by 85%.
  - **Throughput:** Leveraged Python asynchronous workers to unlock a 5x increase in system-wide processing throughput.

## Technical Skills Inventory

- **Machine Learning & Deep Learning (ML/DL):** PyTorch, Hugging Face Transformers, Scikit-learn, Neural Network Architectures.
- **Machine Learning Operations (MLOps):** Amazon Web Services (AWS), Docker Containerization, Kubernetes Orchestration (EKS), MLflow Lifecycle Tracking, GitHub Actions CI/CD.
- **Large Language Model (LLM) Engineering:** Retrieval-Augmented Generation (RAG), Vector Embeddings, LangChain, OpenAI APIs, Prompt Engineering.
- **Programming Languages:** Production Python, Complex SQL Queries.
- **Databases & Vector Stores:** PostgreSQL, MongoDB, Qdrant Vector Database.

## Competitive Engineering Achievements

- **Achievement 1:** Secured Rank 60 in the national Amazon ML Challenge 2025 out of more than 200 competing specialized engineering teams.
- **Achievement 2:** Concluded as the official Runner-Up in the global Supabase Hackathon, out-building a competitive pool of over 200 technical project teams.